BDAG 2: perfection

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Abstract

TODO

Theorem 0.1. *The expected chance of a perfect game with r standard dice of s sides each, given $p := \frac{1}{s}$, is $P(r) = \Sigma_{k=1}^{r} \binom{r}{k} p^k q^{r-k} P(r-k)$, with $P(0)$ defined as 1.*

Proof: This a clear application of the binomial theorem.

Bounding this quantity will be the principal thrust of this paper.

Lemma 0.2 (Easy Cases Lemma)**.** *If $p = 0$, $P(r) = 0$. If $p = 1, P(r) = 1$. If $s = 2$, for any $r > 0, P(r) = \frac{1}{2}$.*

Proof: Though we won't be flipping infinite ($p = 0$) or 1-sided ($p = 1$), dice, these algebraic identities are obvious (can't succeed, can't fail, respectively) and will be useful later.

Base case for s=2: $P(1) = \frac{1}{2}$ since a perfect game occurs if and only if the coin flips 2 instead of 1.

Inductive case for s=2: Assuming $P(k) = \frac{1}{2}$ for all $0 < k < r$, and $P(0) = 1$:

$$\Sigma_{k=1}^{r-1} \binom{r}{k} p^k q^{r-k} P(r-k) + \binom{r}{r} p^r q^0 P(0) = \tag{1}$$

$$(2^r - 2)\frac{1}{2^r}(\frac{1}{2}) + \frac{1}{2^r} 2(\frac{1}{2}) = \tag{2}$$

$$(2^r)\frac{1}{2^r}(\frac{1}{2}) = \frac{1}{2} \tag{3}$$

Theorem 0.3. *For an unbounded set of $r > 0$ standard s-sided dice, $s \geq 2$, the probability $P(r)$ of getting a perfect game has a lower bound of $\exp(-\frac{q}{(1-q)^2})$, if we define $q := 1 - \frac{1}{s}$.*

Proof:

- First, while at stage k, where k dice remain, the probabilty $P_s(k)$ of continuing to some stage $j < k$ is clearly $1 - q^k$.

- The probability $P(k)$ of passing all such stages is greater than or equal to $P^* = \Pi_{i=1}^{k} P_s(k)$, since a successful traversal may involve skipping one or more stages, and P^* assumes we no stages.
- Therefore, $P(k) \geq \Pi_{i=1}^{k} P_s(k) = \Pi_{i=1}^{k}(1 - q^k)$.
- This means $\log(P(k)) \geq \Sigma_{i=1}^{k} \log(P_s(k)) = \Sigma_{i=1}^{k} \log(1 - q^k)$.

This $P(k)$ is apparently defined as the **q-Pochhammer symbol** $(q;q)_\infty = \Pi_{i=1}^{k}(1 - q^k)$, with known convergence properties. I'll instead attempt to tackle convergence without the aid of these theorems.

Often, $\log(1 - z)$ is approximated by $-z$ near zero, but we are most concerned with z near one, since $1 - z$ approaches 0, and $\log(1 - z)$ explodes. Therefore, let's look at the expansion of $\log(1 - z)$:

- Identity 1: $f(z) = -\frac{1}{1-z} = -1 - z - z^2 - \dots$
- Identity 2: $\int f(z) = \log(1 - z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \dots$

So when summing $\Sigma_{i=1}^{k} \log(1 - q^k)$, we can instead sum

$$\log(P(k)) = \log(1 - q^1) + \log(1 - q^2) + \log(1 - q^3) + \dots \tag{4}$$

$$\log(P(k)) = (-q - \frac{q^2}{2} - \frac{q^3}{3} + \dots) + (-q^2 - \frac{q^4}{2} - \frac{q^6}{3} + \dots) + (-q^3 - \frac{q^6}{2} - \frac{q^9}{3} + \dots) + \dots \tag{5}$$

$$> (-q - q^2 - q^3 + \dots) + (-q^2 - q^4 - q^6 + \dots) + (-q^3 - q^6 - q^9 + \dots) + \dots \tag{6}$$

$$= \frac{-q}{1-q} + \frac{-q^2}{1-q^2} + \frac{-q^3}{1-q^3} + \dots \tag{7}$$

$$> \frac{-q}{1-q} + \frac{-q^2}{1-q} + \frac{-q^3}{1-q} + \dots \tag{8}$$

$$= \frac{1}{1-q}[-q - q^2 - q^3 - \dots] \tag{9}$$

$$= \frac{-q}{(1-q)^2} \tag{10}$$

- (2) follows from (1) by substitution using Identity1 above.

- (3) follows from (2) since the denominators of (2) are larger.
- (4) follows from (3) when dividing each sequence by $-q^k$ and applying identity 2.
- (5) follows from (4) since the denominators of (4) are larger.
- (6) and (7) follow from (5) by dividing out $\frac{1}{1-q}$ and applying Identity 1.

Since $\log(P(k)) > \frac{-q}{(1-q)^2}$, $P(k) > \exp(\frac{-q}{(1-q)^2})$.

s	p	q	bound
2	$\frac{1}{2}$	$\frac{1}{2}$	$\exp(-2) \approx .135$
3	$\frac{1}{3}$	$\frac{2}{3}$	$\exp(-6) \approx .00248$
4	$\frac{1}{4}$	$\frac{3}{4}$	$\exp(-12) \approx 6.14 \times 10^{-6}$
10	$\frac{1}{10}$	$\frac{9}{10}$	$\exp(-90) \approx 8.19 \times 10^{-40}$

Table 1: Very loose lower bounds on $P(k = \infty)$

This is a thoroughly unsatisfying lower bound, as we know $P(2) = \frac{1}{2}$.

However, this does provide a positive lower bound, and brings up this interesting fact: **Though increasing the number of sides of the dice can drive P(k) to zero, increasing k to ∞ will never set P(k) to zero**.

If we can couple this with the proof of the next theorem, we can state that $P(k)$ has an asymptotic limit as $k \to \infty$ given some q.

Calculating by hand:

- $P(0) = 1$
- $P(1) = p$
- $P(2) = p^2(1 + 2q) = p^2(3 - 2p)$
- $P(3) = p^3(13 - 27p + 21p^2 - 6p^3)$
- $P(4) = p^4(75 - 316p + 606p^2 - 664p^3 + 432p^4 - 156p^5 + 24p^6)$
- $P(1) - P(0) = p - 1 = -(1 - p)$
- $P(2) - P(1) = p^2(3 - 2p) - p = (-1)p(1 - 2p)(1 - p)$
- $P(3) - P(2) = (-1)3p^2(1 - p)^3(1 - 2p)$
- $P(4) - P(3) = -p^3(1 - 2p)(p - 1)^2(12p^4 - 48p^3 + 72p^2 - 50p + 13)$

Proof 2: $P(t) < P(t - 1)$

- $R(t)$ is starting at position t, throwing $t - 1$ dice.

- $P(t) = p(1 + q^{t-1})P(t-1) + qR(t) = pP(t-1) + pq^{t-1}P(t-1) + qR(t)$
- If $qR(t) < qP(t-1) - pq^{t-1}P(t-1)$, same as $P(t-1) - R(t) > pq^{t-2}P(t-1)$, then $P(t) < P(t-1)$.
- In the special case $q = p = \frac{1}{2}$, then $P(t-1) - R(t) = p^{t-1}(1) - p^{t-1} \cdot p$ + other zero factors $= p^t = pq^{t-2} \cdot \frac{1}{2}$.
- So we're satisfied since $P(t) = \frac{1}{2}$ for $t \geq 1$.
- If $p \leq \frac{1}{3}$, $P(t-1) - R(t) = [p^{t-1}P(0) - p^{t-1}P(1) +$ other positive factors $] = p^{t-1}q + \epsilon$.

$$t = 3, P(3) - R(4) > pq^{4-1}P(4-1) \tag{11}$$
$$p^3P(0) - p^3P(1) \tag{12}$$
$$\binom{3}{2}p^2qP(1) - \binom{3}{2}p^2qP(2) \tag{13}$$
$$\binom{3}{1}p^2qP(2) - \binom{3}{2}p^2qP(3) \tag{14}$$
$$\tag{15}$$